

Manush Prajwal

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Professional Summary

Software engineer with expertise in distributed systems, AI-integrated applications, and cloud infrastructure using NestJS, Docker, PostgreSQL, and AWS. **5X Hackathon Winner** (Google, BNMIT, RNSIT, TOCE) and author of **11 research publications (9 IEEE conference papers and 2 journal papers)** across AI and ML systems, biosensor technologies. Founded **Atlas Bill**, a supermarket ERP platform currently used by **26 stores across Karnataka and Tamil Nadu**.

Education

The Oxford College of Engineering

Sept 2023 – Present

B.E. in Computer Science and Engineering (CGPA: 8.5/10)

Alvas PU College, Moodubidri

2021 – 2023

Pre-University Course in PCMC (Percentage: 91.6%)

Skills

Backend & Systems: Node.js, NestJS, REST APIs, Microservices, Distributed Systems, AWS, GCP

AI & Agentic Systems: LangChain, LangGraph, PhiData, LangFlow, LLM Orchestration, Agentic Loops

DevOps & Infrastructure: Docker, Terraform, Prometheus, Sentry, CI/CD

Databases: PostgreSQL, Supabase, MongoDB

Languages: Python, TypeScript, JavaScript, SQL, C, C++

Achievements

- **5X Hackathon Winner** — Google, BNMIT, RNSIT, TOCE [Certificates | IEEE Xplore](#)
- **3rd Runner-Up** — **Google GenAI Exchange Hackathon 2025** among **278K+ developers** and **4,457 prototypes** submitted in a national-scale generative AI hackathon.
- **Research Publications** Author of **11 research papers (9 IEEE conference papers and 2 journal papers)** across machine learning, distributed systems, and secure computing architectures.
- **Built Atlas Bill ERP platform**, generating **INR 104K revenue in 47 days of launch** with **26+ stores** onboarded.

Work Experience

Freelance Developer

[GitHub](#)

Car Showroom Management Software

May 2023 – Present

- Engineered comprehensive ERP workflows for inventory, bookings, billing, and audit operations, reducing manual showroom operations and processing **INR 320K+ revenue monthly**.
- Designed and optimized transactional database schemas in PostgreSQL handling concurrent inventory and billing operations with **1200+ product SKUs**.
- Implemented RESTful APIs supporting role-based access control, reducing data entry errors.

Projects

AWS CloudPilot — Autonomous FinOps & Infrastructure Optimizer

[Live | GitHub](#)

- Built an autonomous agentic system that monitors cloud infrastructure, detects cost anomalies, and executes safe, deterministic Terraform code reducing potential cloud waste by an estimated **30-40%**
- Architected a 4-stage **Agentic Loop: Monitoring Agent** (parses CloudWatch / Prometheus metrics), **Reasoning Agent** (evaluates guardrails), **Planning Agent** (generates Terraform scripts), and **Execution Agent** (sandboxed deployment).
- Implemented cost anomaly detection (e.g., identifying idle staging clusters costing **\$120/day**) and automated right-sizing recommendations using **Terraform** and AWS SDK integrations, reducing cloud spend by an **estimated 30-40%** per resource.
- Built an agentic Terraform execution environment with validation sandbox requiring no human-in-the-loop, securely identifying and gracefully downsizing idle instances.

Atlas Bill — Retail ERP Startup Platform

[Live | GitHub](#)

- Founded and scaled a retail ERP platform now powering daily operations for **26 supermarkets across Karnataka and Tamil Nadu**, generating **INR 104K revenue in 47 days of launch**.
- Developed comprehensive ERP modules for billing, inventory, CRM, analytics, audit tracking, and financial reporting, reducing manual operational effort by **65%**.
- Engineered real-time inventory synchronization across **26 stores** with automated billing workflows, processing **600 K + monthly transactions** with **99%** reliability.
- Architected backend infrastructure using **Node.js, PostgreSQL, Docker, Built a multi-tenant database architecture** supporting high-scale business transactions.
- Managed billing, inventory, and analytics workflows across thousands of daily retail transactions with **26 concurrent store managers**.